A Data Science Perspective on Indian Government Welfare Schemes:

A Systematic Literature Review and Analytical Framework

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Abstract—Indian government welfare schemes constitute one of the world's largest public policy ecosystems, disbursing hundreds of thousands of crores annually across agriculture, health, housing, employment, education, and financial inclusion. This paper presents a systematic literature review of 18 peer-reviewed and institutional research papers on major central government schemes, synthesised through a data science lens. We categorise findings across four thematic clusters: (i) multi-scheme macro-analyses; (ii) agricultural income support schemes (MGNREGA, PM-KISAN, PMFBY); (iii) health insurance schemes (Ayushman Bharat–PMJAY); and (iv) fiscal sustainability and subsidy rationalisation. For each cluster, key data sources, methodological patterns, performance metrics, and research gaps are identified. The review surfaces recurring challenges in beneficiary targeting, out-of-pocket expenditure reduction, regional disparities, and digital implementation. A structured 1,500-row benchmark dataset is proposed for data science research spanning data cleaning, exploratory analysis, predictive modelling, and visualisation.

Keywords—government schemes, welfare policy, PM-KISAN, MGNREGA, Ayushman Bharat, Direct Benefit Transfer, data science, systematic review, India, social protection, open government data

I. INTRODUCTION

India's central government currently administers over 700 Centrally Sponsored Schemes (CSS) and Central Sector Schemes spanning virtually every domain of human development [1]. Social services expenditure has grown at a compounded annual rate of 15%, rising from ₹14.8 lakh crore in FY21 to a projected ₹25.7 lakh crore in FY25 [4]. Yet scale alone does not guarantee impact: implementation quality, leakage prevention, and beneficiary targeting remain perennial challenges across the scheme landscape.

The emergence of data science as a discipline offers transformative tools for analysing scheme performance. Large-scale administrative datasets — beneficiary records, disbursement logs, grievance registries, and MIS portal exports — are increasingly available through the Open Government Data Platform (data.gov.in), the National Data and Analytics Platform (NDAP, ndap.niti.gov.in), and scheme-specific portals. These datasets, however, suffer from structural issues including missing values, inconsistent categorical formats, duplication, and pronounced regional heterogeneity — precisely the conditions under which data science methods deliver the most value for policy research.

This paper makes three principal contributions. First, it synthesises 18 peer-reviewed and institutional studies on Indian welfare schemes published between 2014 and 2025 [1]–[18]. Second, it maps the methodological approaches deployed across this literature, identifying quantitative, qualitative, and mixed-methods traditions and their relative strengths. Third, it proposes a structured 1,500-row research dataset as a benchmark corpus for data science projects, with 24 feature columns designed to support data cleaning, exploratory data analysis (EDA), classification, regression, and visualisation exercises across the full data science lifecycle.

II. METHODOLOGY

This review follows a structured literature synthesis protocol. Papers were identified through four databases: ResearchGate, PubMed, Google Scholar, and institutional repositories including 3ie, NIPFP, and the Centre for Policy Research (CPR India). Search terms combined scheme names (PM-KISAN, MGNREGA, PMJAY, DBT), evaluation terms (impact, effectiveness, utilisation, coverage), and methodological descriptors (survey, cross-sectional, administrative data). The time range spanned 1995–2025, with deliberate emphasis on post-2018 studies to capture the current policy generation.

Inclusion criteria required that papers (i) focus on a named Indian government scheme, (ii) present empirical or analytical findings rather than purely descriptive accounts, and (iii) be traceable to a verifiable institutional or peer-reviewed journal source. Eighteen papers satisfied these criteria and are organised into four thematic clusters. Table I provides a consolidated reference summary across all reviewed works.

TABLE I. BENCHMARK DATASET FEATURE DICTIONARY (1,500 ROWS × 24 COLUMNS)

Feature Name	Data Type	Missing %	Unique Values	Range / Sample Values	Role in Analysis
Scheme_ID	Text	0.0%	1,500	S0001 – S1500	*Primary Key / Identifier*
Scheme_Name	Text	0.0%	1,467	PM Jan Dhan Yojana ...	*NLP / Name Clustering*
Ministry	Categorical	0.0%	46	Min. of Finance, Health ...	*Grouping / Pivot*
Category	Categorical	0.0%	41	Health, Agriculture, Housing ...	*Classification Feature*
Launch_Year	Numeric	0.0%	73	1952 – 2024	*Time-Series / Decade EDA*
Target_Beneficiaries	Categorical	0.0%	27	Rural Poor, Farmers, Youth ...	*Beneficiary Segmentation*
Budget_Allocated_Crore	Numeric	8.2%	1,376	85 – 4,99,455 Cr	*Regression / Imputation*
Budget_Utilized_Crore	Numeric	18.5%	1,222	59 – 5,12,055 Cr	*Key Missing — Imputation*
States_Covered	Text/Mixed	0.0%	37	'All States' or 1–36	*Spatial / Cleaning Task*

Feature Name	Data Type	Missing %	Unique Values	Range / Sample Values	Role in Analysis
Beneficiaries_Reached_Lakh	Numeric	**15.7%**	1,265	0.003 – 14,959 Lakh	*Impact Metric / EDA*
Status	Categorical	0.0%	4	Active / Completed / Closed ...	*Classification Label*
Mode	Categorical	0.0%	24	DBT / Subsidy / Grant / Loan ...	*Feature Encoding*
Gender_Focus	Categorical	0.0%	6	All / Women / Children / SC-ST	*Equity Analysis*
Age_Group	Categorical	0.0%	21	All Ages / 18+ / 0-6 / 60+ ...	*Beneficiary Sub-grouping*
Application_Mode	Categorical	**20.7%**	4	Online / Offline / N/A ...	*Imputation / One-Hot Enc.*
Success_Rate_%	Numeric	6.1%	607	30.0% – 100.0%	*Primary Regression Target*
Grievances_Filed	Numeric	0.1%	1,496	397 – 4,99,508	*Governance Metric*
Grievances_Resolved	Numeric	5.4%	1,419	264 – 4,85,924	*Resolution Rate Feature*
Scheme_Duration_Years	Text/Mixed	0.0%	86	1–78 or 'Ongoing'	*Parsing / Type Cleaning*
Annual_Report_Available	Categorical	0.0%	2	Yes / No	*Boolean Feature*
Website_Available	Categorical	0.0%	2	Yes / No	*Boolean Feature*
PM_Flagship	Categorical	0.0%	2	Yes (18.7%) / No (81.3%)	*Binary Classification*
Nodal_Agency	Text	0.0%	54	NABARD, NHM, NSDC, DPIIT ...	*Agency-Level Grouping*
Remarks	Text	**25.5%**	18	'Missing data' / 'Merged'	*Annotation / NLP*

Bold red = missing >20%; Orange = missing >10% — both require imputation. Alternating row shading for readability. Dataset spans 46 Ministries, 41 Categories, Launch Years 1952–2024. DBT = Direct Benefit Transfer; EDA = Exploratory Data Analysis; NLP = Natural Language Processing.

III. CLUSTER A: MULTI-SCHEME MACRO-ANALYSES

A. Effectiveness and Economic Impact

Chaudhary [1] provides one of the most comprehensive single-paper treatments of Indian welfare program effectiveness, examining the Public Distribution System (PDS), NREGA, and PMJDY through a mixed-methods lens. The study finds that while poverty alleviation metrics show measurable improvement, implementation variability across states significantly dilutes aggregate national outcomes. The paper underscores that digital integration — particularly Aadhaar-linked DBT — has the potential to transform welfare delivery accuracy and reduce leakage at scale.

Jha [2] situates India in a comparative Asia-Pacific context and finds that India's economic growth has consistently outstripped its social welfare performance. Despite ranking among the fastest-growing major economies since 2009, India spends less on social welfare as a share of GDP than comparable nations in the region. Female access to welfare programs emerges as a critical and chronically underserved dimension requiring targeted policy intervention.

B. Two-Decade Spending Trend Analysis

An analysis published in The India Forum [5] compares combined spending on ten major schemes across FY2004–05 and FY2023–24 as a percentage of GDP. The study finds negligible net change in the aggregate welfare share of GDP across the UPA and NDA periods, though the composition shifted markedly: UPA-era public-goods schemes (MGNREGA, ICDS) faced sustained funding squeezes while NDA-era private-goods schemes (housing, LPG, water connections) received proportionally higher allocations. This composition shift has significant implications for long-run human capital accumulation.

The CPR India evolution report [4] supplements this by documenting the transformation of government MIS platforms across 2008–2023, identifying improvements in real-time data availability alongside persistent inter-state variation in data quality, completeness, and standardisation — a critical concern for any quantitative analytical work.

C. Rural Poverty and Scheme Convergence

The NRJBE study [3], drawing on 2023–2025 data, evaluates eight major rural schemes simultaneously and concludes that MGNREGA and NRLM strengthened livelihood resilience and women's empowerment, while PMGKAY and PM-KISAN ensured food and income security during and after COVID-19. Structural challenges including payment delays, inclusion errors, fiscal limitations, and pronounced inter-state variation remain pan-scheme concerns requiring systemic redress.

IV. CLUSTER B: AGRICULTURAL INCOME AND EMPLOYMENT SCHEMES

A. MGNREGA — Employment Guarantee Act

The 3ie Working Paper [6] represents the most rigorous systematic evidence synthesis on MGNREGA available in the public domain. Covering studies published between 2006 and 2015 across all Indian states, it maps outcomes across economic categories (wages, consumption, agricultural productivity), empowerment (women's participation, child labour reduction), and governance (fund utilisation, social audit compliance). The review identifies a consistent positive wage effect but notes substantial heterogeneity in asset creation quality and governance outcomes across states, reflecting the federated nature of scheme implementation.

The DBT-MGNREGA analysis [7] examines the integration of Aadhaar-based payment infrastructure into the scheme's Electronic Payment System, developed in February 2015. The paper notes that while DBT integration has materially improved payment transparency and reduced wage delays, challenges remain in Aadhaar-seeding at the gram panchayat level, particularly in tribal, hilly, and remote geographies where connectivity is limited.

B. PM-KISAN — Direct Income Support

Varshney et al. [9] establish a causal linkage between PM-KISAN income transfers and adoption of modern agricultural inputs using data from the Economic and Political Weekly dataset. The study demonstrates that the ₹6,000 annual transfer, while modest in absolute terms, reduces the credit constraint for marginal farmers and directly facilitates investment decisions in seeds, fertilisers, and crop protection that would otherwise be deferred until post-harvest liquidity is available.

Bordoloi et al. [10] examine the scheme's differential impact in the North-Eastern states, where baseline agricultural productivity was historically below the national average. They report record growth in food grain and oilseed production following PM-KISAN implementation, attributing this to a compounding multiplier effect of income support on input expenditure across multiple crop cycles.

A micro-level farm survey in Ri-Bhoi district, Meghalaya [11] quantifies the scheme's effect on paddy cultivation economics at the household level: beneficiary farmers recorded a 2.14% rise in cost of cultivation (reflecting higher input adoption), accompanied by 4.74% higher gross returns and 8.71% higher net returns compared to matched non-beneficiary households. This micro-level evidence triangulates the aggregate econometric findings of Varshney et al. at village scale.

A statistical study covering beneficiary distribution from 2019 to 2023 [8] employs the Kruskal-Wallis non-parametric test to reveal statistically significant regional disparities. Uttar Pradesh consistently dominated beneficiary counts, while the 2022–23 decline in registered farmers across several states is attributed to stricter Aadhaar-linked land record validation requirements introduced to reduce ineligible registrations.

V. CLUSTER C: AYUSHMAN BHARAT – PMJAY

A. National Coverage and Financial Protection

The most methodologically rigorous assessment of PMJAY's impact to date is a two-wave household survey conducted in Chhattisgarh [12], covering approximately 15,000 individuals across 2021 and 2022. The study finds that four years post-launch, PMJAY has achieved large population coverage but has not demonstrably reduced out-of-pocket expenditure: among PMJAY-enrolled beneficiaries hospitalised at private facilities, catastrophic expenditure rates stood at 78.1% in 2021 and 70.9% in 2022. The study attributes this to continued overcharging by private hospitals contracted under the scheme and ineffective purchasing mechanisms.

B. State-Level Utilisation Studies

Sriee and Maiya [13] conducted a cross-sectional survey among 300 households in Tamil Nadu. Coverage was 42.33%, and among covered households only 47.24% had actually utilised scheme benefits in the prior year. The study recommends intensive ASHA and Anganwadi worker training as the primary lever for improving community-level scheme linkage and effective utilisation.

A parallel multi-stage sampling study in Gujarat [14] across 1,152 households in three districts finds a utilisation rate of 43.3%. Despite high awareness and enrolment, significant out-of-pocket costs persist — largely because beneficiaries seek care at non-empanelled facilities or incur informal charges within empanelled ones, eroding the intended financial protection benefit of the scheme.

Yadav and Bora [15] apply a data analytics lens to PMJAY evaluation by analysing district-level transaction data from Assam's DHIS platform. A Pareto analysis reveals that Kamrup district accounts for a disproportionate share of all hospital transactions in the state, indicating extreme concentration of healthcare utilisation in and around the state capital and raising serious equity concerns for rural and remote districts.

C. Portability and Region-Specific Challenges

The portability study [16] examines PMJAY's cross-state treatment feature in Tamil Nadu. Incoming portability admissions generated ₹1.85 crore in claims while outgoing admissions generated ₹84 lakh,

indicating net inflows of complex cases into better-equipped tertiary facilities. A critical process gap is identified: approximately 40% of claim payments exceeded the mandated 30-day processing window, creating cash-flow stress for empanelled hospitals and potentially discouraging participation.

Evaluating PMJAY in the Kashmir Valley [17] adds a dimension frequently absent from national studies: the intersection of geographic remoteness, socio-political complexity, and healthcare infrastructure gaps. The mixed-methods study finds that conventional Information-Education-Communication (IEC) strategies are insufficient for this context and calls for region-specific awareness campaigns, infrastructure investment, and capacity-building among local healthcare providers.

VI. CLUSTER D: FISCAL SUSTAINABILITY AND STATE SUBSIDIES

Amarnath et al. [18], drawing on budgetary data from seven states — Punjab, Rajasthan, Uttar Pradesh, Odisha, Andhra Pradesh, West Bengal, and Nagaland — over 2018–19 to 2022–23, present a stark picture of fiscal stress at the sub-national level. States with persistent revenue deficits are spending over 20% of their revenue expenditure on subsidies, while fiscal space has contracted below 50% in several cases. The study calls for disciplined subsidy rationalisation strategies aligned with international best practices in targeted subsidy design and enforcement.

From a data science perspective, this cluster directly informs the structural design of the benchmark dataset. The dataset encodes budget allocation and utilisation figures reflecting these fiscal pressures, with systematic missingness in the Budget_Utilized_Crore field (approximately 18.5% missing) providing a realistic imputation challenge. The relationship between fiscal stress indicators and scheme success rates offers a compelling regression target for supervised learning exercises.

VII. DATA SCIENCE RESEARCH IMPLICATIONS

A. Benchmark Dataset Design

The 1,500-row benchmark dataset spans 46 ministries, 41 scheme categories, and 72 years of launch history (1952–2024), with 24 feature columns. Key numeric features include Budget_Allocated_Crore (8.2% missing), Budget_Utilized_Crore (18.5% missing), Beneficiaries_Reached_Lakh (15.7% missing), Success_Rate_% (6.1% missing), Grievances_Filed, and Grievances_Resolved. Categorical features include Status (Active/Completed/Closed/Under Review), Mode (17 delivery types), Gender_Focus, Age_Group, and Application_Mode. The structured missingness pattern mirrors real-world administrative data and provides non-trivial imputation challenges suitable for graduate-level practicum.

B. Analytical Tasks Enabled

- Data Cleaning: Standardise Mode, Status, and Age_Group; parse Scheme_Duration_Years (mixed numeric and string); handle Budget zero-values for contributory schemes
- EDA: Distribution analysis by ministry, category, launch decade; correlation matrix between budget allocation, utilisation rate, and success rate
- Predictive Modelling: Regress Success_Rate_% on financial, governance, and structural features; classify Active vs Closed/Completed using ensemble methods
- Visualisation: Ministry-wise budget utilisation heatmaps; time-series of scheme launches by decade; grievance resolution funnel by category
- NLP / Clustering: Named entity recognition on scheme names; K-means clustering by combined financial and performance features

C. Alignment with Literature Gaps

The reviewed literature consistently identifies three data infrastructure gaps that the benchmark dataset is designed to address: (1) absence of a unified cross-scheme comparison corpus enabling multi-scheme analysis [2, 4]; (2) lack of granular financial and temporal features linking scheme design parameters to performance outcomes [5, 18]; and (3) insufficient open-access infrastructure for replicable, reproducible analytical work [3, 6]. The benchmark dataset, available in CSV format, is designed as a direct response to these gaps.

VIII. CONCLUSION

This paper has presented a systematic review of 18 empirical and analytical studies on Indian government welfare schemes, organised across four thematic clusters: macro welfare analyses, agricultural income and employment, health insurance, and fiscal sustainability. A consistent pattern emerges across all clusters: India's schemes achieve significant reach in terms of registered beneficiaries and fund disbursement, but face persistent gaps in actual utilisation, financial protection efficacy, and equitable geographic distribution — particularly between urban cores and rural peripheries.

For the data science research community, this literature surfaces a rich, practically important, and methodologically underexplored domain. The combination of large administrative datasets, structural data quality issues, multi-level federated governance, and high-stakes societal outcomes makes government scheme data an ideal corpus for advancing methods in imputation, causal inference, geospatial analysis, and responsible artificial intelligence. The 1,500-row benchmark dataset accompanying this review provides a concrete, documented starting point.

Future work should prioritise live integration of scheme data from official portals (nrega.nic.in, pmkisan.gov.in, nha.gov.in, data.gov.in) with the static benchmark, enabling real-time monitoring dashboards and predictive early-warning systems for scheme underperformance. Collaboration between the data science community, policy researchers, and government agencies will be essential to realise this potential at scale.

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